

Mark IV Planetary Exploration Spacesuit Pressure Garment Maintenance

1. Pressure Garment Assembly

The Mark IV suit operates at 29.6 kPa (4.3 psi) pure oxygen atmosphere, featuring rear-entry composite upper torso and articulating titanium rotary bearing joints.

2. Portable Life Support System (PLSS)

PLSS backpack integrates dual lithium hydroxide CO₂ scrubbers, sublimator ice cooling plates, and dual high-pressure oxygen tanks pressurized to 240 bar.

3. Micrometeorite & Dust Layer

Outer orthofabric cover combines Nomex, Kevlar, and aluminized Mylar to defend against abrasive basalt dust particles and micrometeorite impacts up to 5 km/sec.